

SNAPDRAGON AI LAB BUILD & PRESENT CHALLENGE 2026

Sahaay

सहाय · assistance

Hands-free and eyes-free control of a Windows PC, *entirely on the Snapdragon Hexagon NPU.*

Head and voice replace the mouse and keyboard; a conversational narrator replaces the screen. Six AI models, no cloud, no data leaves the laptop.



TWO PEOPLE

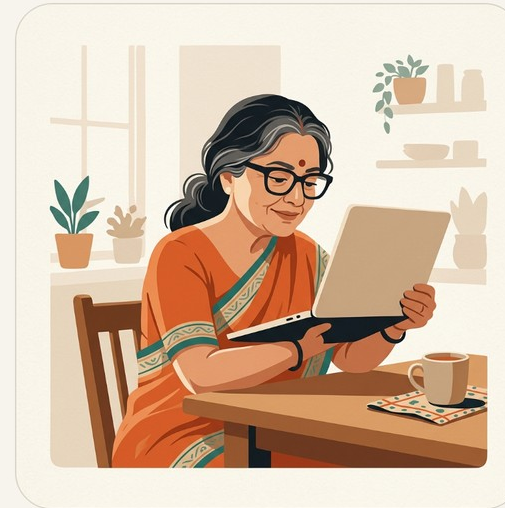
Two groups of people who cannot use a laptop the way it ships.



Arjun, 22

Engineering student · cerebral palsy

Cannot hold a trackpad steady. Every assignment, portal and chat needs **someone else's hands.**



Meena, 58

Shopkeeper · low vision, diabetic retinopathy

Can hear a screen reader list forty buttons. **Cannot see the photo her daughter sent,** or the chart in her bank app.

2.68 crore

Indians living with a disability
Census of India 2011

1 in 5

of them with a movement disability
Census of India 2011

1 in 5

of them with a visual disability
Census of India 2011

Composite personas drawn from published accessibility research; illustrations, not real people.

THE PROBLEM

Assistive technology today is expensive, depends on the cloud, and *stops at the basics*.

COST

₹1.5 lakh+

for an eye-gaze system

Plus a licence, a dongle, and often a subscription. Out of reach for a student on a ₹60,000 laptop budget, and rarely covered by insurance in India.

Tobii Dynavox, PCEye and comparable Indian resellers, 2026 list prices

PRIVACY

24_{h/day}

of face video and screen content

A cloud assistant has to stream the camera and the screen to a server. Face data is **sensitive biometric data under India's DPDP Act 2023**; the screen shows bank statements and medical records.

Digital Personal Data Protection Act 2023, s.2(t) and s.9; typical cloud speech and vision APIs

COVERAGE

0

built-in tools that describe a photo or the room

Screen readers read controls, not images. Voice control types and clicks, but still needs a hand for everything else. **Nothing ties head, voice, screen and camera together.**

Windows Narrator and Voice Access feature scope, Microsoft docs, 2026

THE IDEA

Run the entire assistive stack on the NPU that is *already inside the laptop.*

A Snapdragon Hexagon NPU runs face tracking, speech, a language model and a vision model at the same time, at a few watts, with nothing leaving the machine. HP ships it in the OmniBook that students already buy.

₹1.5 L+

eye-gaze hardware, licence,
cloud account

₹0

Sahaay: a download for the
laptop you already own



Two modes, one on-device brain: *hands-free control* and a *screen narrator*.

HANDS-FREE

Head moves the cursor. Voice types and commands.

For Arjun: motor impairment, injury, RSI

- **Nudge your head**, the cursor moves. Hold still, it stays. Calibrates itself.
- **Blink** to click, **hold still** to click, **open your mouth** to drag.
- **Dictate** into any app, in English or Hindi.
- **Say it:** *"open my downloads", "write a thank-you note to my teacher and save it."*
- **Or press a switch.** Any Arduino becomes a button interface.

NARRATOR

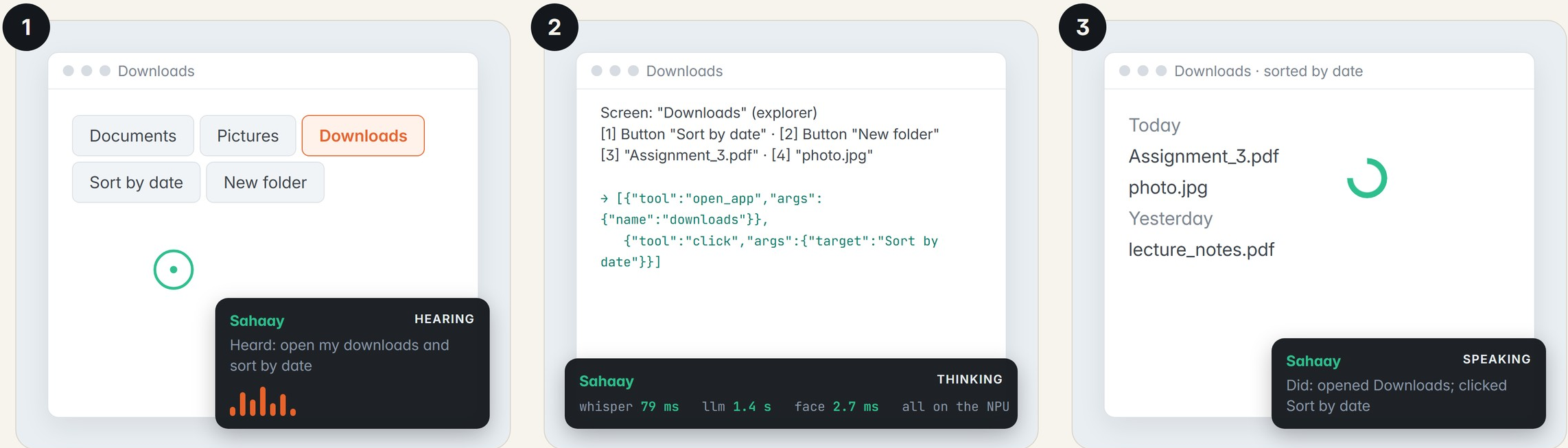
The screen and the camera, described in speech.

For Meena: low vision, blindness

- *"What's on my screen?"* Two spoken sentences: **which app, what content, what matters.**
- *"Read this."* The paragraph under focus, in her language.
- *"Describe the picture."* A vision model explains **photos, charts, canvases.**
- *"What do you see?"* The webcam view, labels and signs, **read aloud.**
- Every action is **confirmed by voice**, so she always knows what just happened.

ONE COMMAND, END TO END

How a spoken request becomes a *validated action* and a spoken confirmation.



ARJUN SAYS

"Open my downloads and sort by date."

ON THE NPU

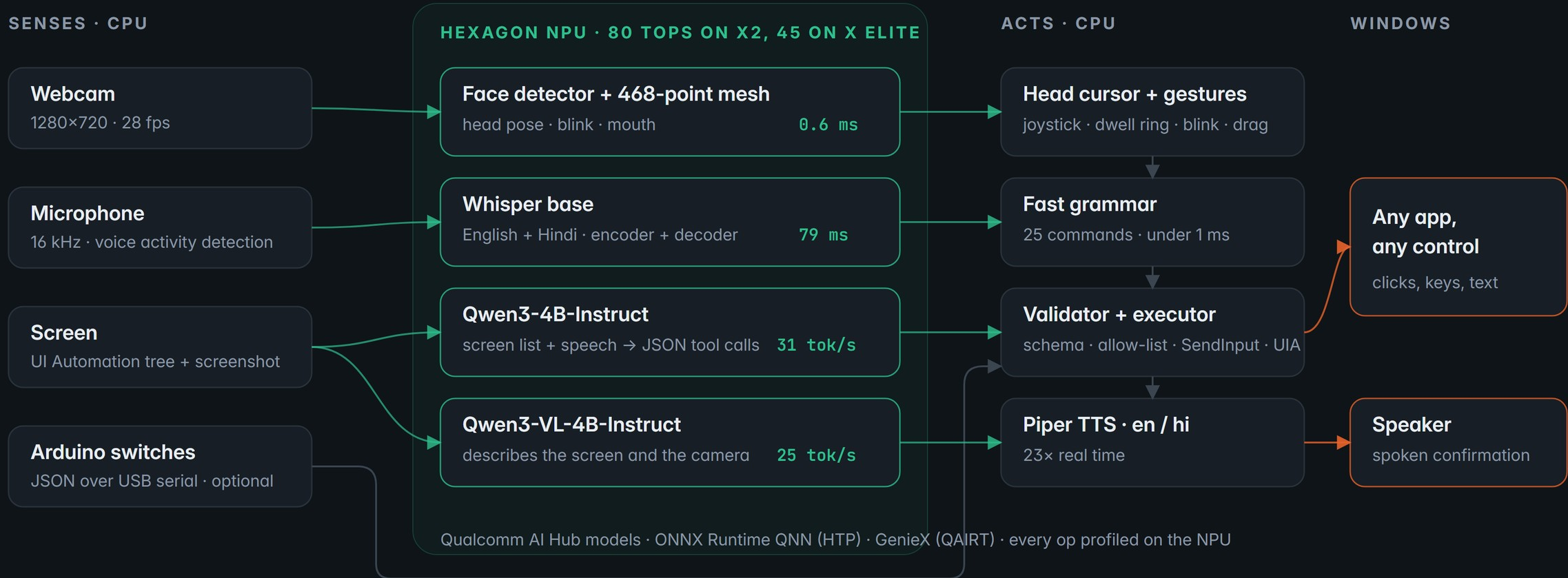
Speech becomes text, the screen becomes a list, the list plus the request becomes two safe tool calls.

SAHAAY SAYS

"Downloads is open, sorted by date. Newest first is Assignment 3."

HOW IT WORKS

Six models on the Hexagon NPU. Capture, validation and input injection on the CPU. *The GPU untouched.*



— neural network, runs on the NPU — plain code, CPU — effect on Windows GPU: untouched, left to the user's apps

MEASURED PERFORMANCE

Every number is measured, and the NPU execution provider is *asserted* before it is reported.

Latency per stage, CPU vs NPU

Median of 30 runs on the development laptop (Snapdragon X2 Elite). Same graph, same inputs.

Face pipeline

detect, crop, 468-pt mesh, pose · live 720p frame



2.7 ms

Face detector

BlazeFace 256×256



0.43 ms

Face mesh

468 landmarks 192×192



0.16 ms

Oryon CPU Hexagon NPU via QNN bars scaled per row

`tools/npu_check.py` asserts the active provider for every model. The usual `providers= [...]` call silently runs on the CPU with today's runtime; Sahaay selects the NPU device explicitly.

Sahaay

Speech and generative models, NPU only

Whisper ships as a precompiled QNN context; the language models run through GenieX on QAIRT.

79 ms

to transcribe a 5.6 s command with Whisper base (21 ms encoder, 2.7 ms per token)

NPU

31 tok/s

Qwen3-4B planning tool calls; 243 ms to first token with the screen in context

NPU

25 tok/s

Qwen3-VL-4B describing the screen or the webcam; 269 ms to first token

NPU

10 s

to load each generative model; face tracking is up in 2 s, so the cursor works almost immediately

NPU

Qualcomm AI Hub, reference devices, every op on the NPU

X2 Elite	face 0.4 ms · mesh 0.2 ms · Whisper enc 21.5 ms · dec 2.4 ms/token
X Elite (HP OmniBook)	face 0.7 ms · mesh 0.3 ms · Whisper enc 45.4 ms · dec 3.7 ms/token

08 · benchmarks/results.md and benchmarks/aihub_profiles.md in the repository

One command installs it on any Snapdragon X or X2 PC.

After that it never needs the internet.

```
# any Snapdragon X / X2 PC · HP OmniBook, EliteBook · no admin rights
powershell -ExecutionPolicy Bypass -File install.ps1
# Python if missing · venv · models for your chipset, once · espeak-ng · Start Menu
shortcut
```



Private by construction

No account, no cloud, no telemetry. Camera frames, audio and screenshots **stay in RAM** and are discarded. A local log file for support, nothing else written.



Fits the OmniBook a judge owns

Hands-free mode within 16 GB. Both generative models resident on 32 GB (5.4 + 6.1 GB measured), or the 2B vision model on 16 GB. 45 TOPS is enough; 80 TOPS has headroom.



Usable without sight or hands

Global hotkeys, a tray icon, a heads-up panel that shows what was heard and done, **spoken confirmation of every action**, English and Hindi voices, and switch access through any Arduino, including the UNO Q from the AI Lab kit.

Where the cloud fails this user, and the *trade-offs* Sahaay accepts in return.

WHAT THE USER NEEDS	CLOUD ASSISTANT	SAHAAY ON THE HEXAGON NPU
Privacy of a face camera and the screen	Frames and screenshots leave the device continuously. Face data is sensitive biometric data under India's DPDP Act 2023 (s.2(t)); the screen carries bank and medical records	Processed in RAM, discarded. zero network calls after install; verifiable with Wi-Fi off
Cursor latency a person can tolerate	A network round trip before any inference. typically 50 to 150 ms to a cloud region from India, more on congested mobile links	2.7 ms per frame for the cursor; 79 ms for a spoken command. measured, benchmarks/results.md
Works in a hostel, a hospital ward, a train	Stops when the link drops. every command is a request; there is no offline mode	Fully offline. 6 GB of models cached once; nothing else fetched
Battery on a laptop that is the only interface	Video upload keeps the radio busy; inference still runs somewhere. continuous 720p upload is 1 to 3 Mbps over Wi-Fi or 4G	The CPU path would cost 25 ms per frame; the NPU does it in 2.7 ms and the CPU stays idle for the user's apps. measured, same frame
Cost to a student	Per-minute billing for speech, plus vision and language calls. Whisper API at \$0.006/min: three hours a day is about ₹2,700 a month, speech alone (₹84/\$)	Zero after the laptop. no subscription, no API keys

TRADE-OFFS WE ACCEPT **4-billion-parameter models** reason less well than frontier cloud models, so common commands bypass them through a fixed grammar and every model action is validated. **Each generative model takes 10 s to load** and 5 to 6 GB of RAM. **Whisper base mishears in noisy rooms**; fragments are dropped rather than acted on.

Seven problems between the models and a working product, *and their fixes.*

01 ONNX Runtime silently ran on the CPU while claiming the NPU

With the current plugin packaging, `providers=["QNNExecutionProvider"]` is accepted and ignored. **Fix:** register the QNN plugin, pick the NPU device through the device API, assert the active provider on every session, and ship `npu_check.py` so a reviewer can see it.

02 No OpenCV for Windows on ARM64

The camera and every image operation had to work without it. **Fix:** DirectShow through COM for capture (with `CoInitialize` on every worker thread), Pillow and numpy for the MediaPipe letterbox, affine crop and landmark mapping.

03 Whisper ships as a precompiled QNN context with no Python runner

Qualcomm's reference decode loop needs torch. **Fix:** re-implemented the encoder call and the 200-step KV-cache decoder loop in numpy: six layers of self-attention cache, causal mask and forced language prefix.

04 Piper's phonemiser is not built for ARM64

The wheel's espeak bridge fails to import. **Fix:** phonemise through the espeak-ng CLI and drive the VITS ONNX graph directly; English and Hindi voices at 23× real time.

05 Two GenieX models cannot initialise at the same time

Concurrent loads fail to create HTP contexts. **Fix:** a single loader thread brings up the language model, then the vision model, while the cursor and speech are already live.

06 A mis-heard fragment launched Windows Search

"Open it." from clipped audio became a search for "it". **Fix:** pronoun and length guards in the grammar, an allow-list of tools and elements, and no search fallback for short tokens.

07 QNN context caching and chipset generations

A cached compiled graph must be reused, not regenerated, and the development laptop's Hexagon v81 binaries do not run on the OmniBook's v73. **Fix:** reuse the cache when present, invalidate on mismatch, and ship separate model packs per chipset, both profiled on Qualcomm AI Hub.

ROADMAP

What ships next, in order.

- **Gaze on the NPU.** The AI Hub eyegaze model for coarse screen-region selection alongside head pose, for people with very limited neck movement.
- **Piper on the NPU, Hindi and Hinglish commands.** The AI Hub Piper recipe, and a command grammar that understands "neechе scroll karo".
- **Grounding from the vision model.** "Click the blue button" when an app has no accessibility tree: games, canvases, remote desktops.
- **Trials with an assistive-technology centre.** Calibration profiles for tremor and limited range.
- **MSIX package** for the Microsoft Store, and OmniBook presence-sensor auto-pause.

A ₹60,000 laptop that *anyone* can use, because the NPU is already inside it.

Everything in this deck is in the repository: code, benchmarks, installer, model packs and the reviewer tools that prove the NPU is doing the work.

github.com/Vision-jarvis/sahaay

Ruhan Srivastava

IIT Kharagpur · built on a Snapdragon X2 Elite with Qualcomm AI Hub, ONNX Runtime QNN and GenieX

